Advertising, Social Media and Parasocial Recommendations

New Infuencer-Centered Modeling Approach

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Model Setup: The Product Landscape

- Multiple product categories

  $\Rightarrow$ denote $k \in \{A, B, \dots\}$

  $\Rightarrow$ will allow for influencer "specialization"

  $\Rightarrow$ relevant only for informational purposes

  Not yet used right now

- Multiple quality dimensions

  $\Rightarrow$ denote $j \in \{1, 2, \dots\}$

  $\Rightarrow$ relevant for realized utility

Model Setup: The Firm Side

The Monopolist

- Single firm with high (h) or low (l) quality in each dimension [generically q_j]
- Realization of all q_j are private information with $P[q_j = H] \equiv \pi$
- Fix $|J| = 2$ gives three classes q: H (both $q_j = h$), M (one $q_j = h$) or L (no $q_j = h$)
- Produced at constant, homogeneous marginal cost c
- Competes (indirectly) against outside option
- Can advertise product on social media at exogenous cost κ

The Outside Option

- Available instead of purchasing the product

$\Rightarrow$ Can be thought of as background Bertrand competition with cost $c_0 = v_0 + p_0$

Model Setup: The Consumer Side

- One (representative) consumer with unit demand
- Randomly drawn product dimension type $\theta \in J$
- Receives utility v_0 from outside option and v_q from the new product

  $\Rightarrow$ utility of product given by $u_c(\theta) = v_{q_\theta}$ [i.e. quality of dimension θ]

  $\Rightarrow$ normalize $v_l = 0$, simplify notation $v_h = v$

- Can obtain **one** recommendation r prior to purchase

  $\Rightarrow$ either from environment (non-strategic) or social media (potent. strategic)

Assumption 1

The outside option v_0 is sufficiently large to dominate random purchase.
That is, $v_0 > \pi v - p$.

Model Setup: The Influencer Side

- One (representative) Influencer that can give recommendations
  - Product dimension type $\iota \in J$
  - direct utility from payment $\Pi_{\mathcal{I}}$, indirect utility from value-added to consumer

    $\Rightarrow$ altruism parameter α captures trade-off, distr. following (cont.) F_α

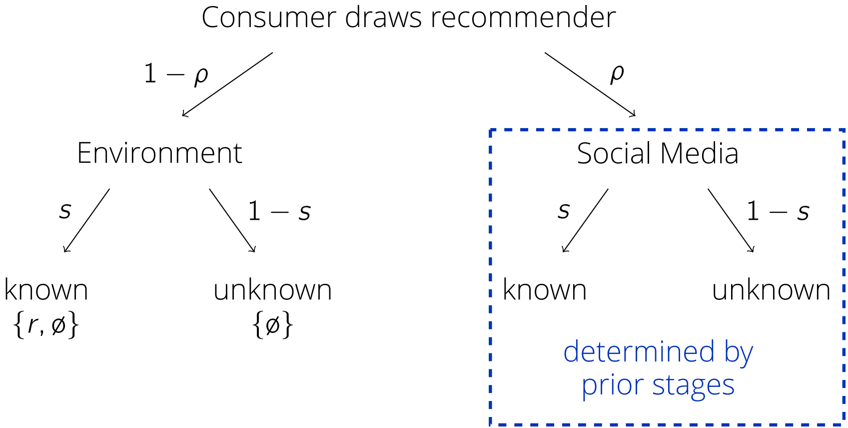
    $$u_{\mathcal{I}} = (1-\alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - u_c(\emptyset)] = (1-\alpha)\Pi_{\mathcal{I}} + \alpha[\mathbb{E}[v_\theta|r] - p - v_0]$$

  $\Rightarrow$ non-strategic environment encapsuled by $\alpha = 1$
- Type-match probability ("similarity") σ $\Rightarrow$ potentially σ_0, σ_I

Parasocial Relationships

The consumer's parasocial relationship is captured by overestimating the influencer's similarity, i.e., $\sigma_\eta > \sigma$. [Specific form to be determined]

Model Setup: The Recommendation Procedure



Model Setup: Timing

1. Fundamentals are drawn $(p, \kappa, q, v_j, \ldots)$
2. Firms learn q, knowing recommenders learn v_ι
3. Firm decides on endorsement offer
4. Influencer decides on accepting endorsement offer
5. Consumer draws recommender
6. Consumer observers potential recommendation
   $\Rightarrow$ can distinguish genuine from paid recommendations
7. Consumer makes purchase decision

$\Rightarrow$ utilities are realized

An Endorsement-Free Benchmark

- Outside option of v_0 $\Rightarrow$ by assumption, obtained without recommendation
- Consumer updates belief about v_θ after receiving a recommendation: $\pi \xrightarrow{r} \pi_r$

$$\pi_r \equiv \mathbb{P}[v_\theta = v|r] = \sigma \frac{1 \cdot \mathbb{P}[v]}{1 \cdot \mathbb{P}[v] + 0 \cdot \mathbb{P}[0]} + (1-\sigma) \frac{\pi \cdot \mathbb{P}[v]}{\pi \cdot \mathbb{P}[v] + \pi \cdot \mathbb{P}[0]} = \sigma + (1-\sigma)\pi$$

$\Rightarrow$ assume expected utility $\pi_r v - p = \pi v + (1-\pi)\sigma v - p > v_0$

- without endorsements ($\Pi_{\mathcal{I}} = 0$), influencer's interests perfectly aligned

$\Rightarrow$ recommend a (known) product iff $v_\iota = v$

Equilibrium of the Endorsement-Free Benchmark

Without endorsements, there exists an equilibrium in which any high-value product gets recommended and all recommendations are followed.

The HML-Endorsement Equilibrium

HML-Endorsement Equilibrium

For sufficiently small κ there exists an equilibrium with endorsements in which

(a) All firm types ($q = H, M, L$) decide to seek endorsements at cost κ.

(b) There exist thresholds $\bar{\alpha}_s$ and $\bar{\alpha}_{\bar{s}}$ such that an influencer

   i. with $\alpha \leq \bar{\alpha}_{\bar{s}}$ endorses the product unseen.

   ii. with $\alpha \leq \bar{\alpha}_s$ endorses any product and $\alpha > \bar{\alpha}_s$ only $v_\iota = v$ products.

(c) All consumers follow genuine recommendations.

(d) Sufficiently parasocial consumers ($\sigma_\eta > \bar{\sigma}$) also follow endorsements.

- Equilibrium with full firm-type pooling

Firm Decision

- Firm endorses if costs are sufficiently small $\kappa \leq (D_E^q - D_0^q)(p-c)$

  $\Rightarrow$ effective demand with (D_E^q) and without (D_0^q) endorsements depends on v_ι
- Denote share of parasocial customers ($\sigma_\eta > \bar{\sigma}$) by ψ_η

v_ι	D_0	D_E
V	s	$\rho(1-s)\psi_\eta F_\alpha(\bar{\alpha}_{\bar{s}}) - \rho s(1-\psi_\eta) + s$
0	0	$\rho(1-s)\psi_\eta F_\alpha(\bar{\alpha}_{\bar{s}}) + \rho s \psi_\eta F_\alpha(\bar{\alpha}_s)$

- Note that absolute gains ($D_E^q - D_0^q$) decrease with q_ι (and hence with q)

  $\Rightarrow$ all firms profit iff $\kappa \leq \bar{\kappa}_H \equiv \rho[\psi_\eta(1-s)F_\alpha(\bar{\alpha}_{\bar{s}}) - (1-\psi_\eta)s](p-c)$

  $\Rightarrow$ Note: $\bar{\kappa}_H > 0 \iff \frac{\psi_\eta}{1-\psi_\eta} F_\alpha(\bar{\alpha}_{\bar{s}}) > \frac{s}{1-s}$ $\rightarrow$ necessary cond. for HML-eq.

Consumer Decision

- Reminder: Consumers can differentiate paid from genuine recommendations

  $\Rightarrow$ paid ones can still be **aligned** (i.e., iff $v_\iota = v$) for high α $\Rightarrow$ denote prob. ψ_α

- Genuine recommendations always aligned (Benchmark) $\Rightarrow$ always followed

- Follow paid recommendation iff v_0 or p sufficiently small and

$$\mathbb{E}[v_\theta | r^p] = \psi_\alpha(\pi + (1-\pi)\sigma)v + (1-\psi_\alpha)\pi v = [\pi + (1-\pi)\sigma\psi_\alpha]v \geq v_0 + p$$

  $\Rightarrow$ satisfied for sufficiently parasocial consumers $\Rightarrow \sigma_\eta \geq \bar{\sigma} \equiv \frac{v_0+p}{(1-\pi)\psi_\alpha v} - \frac{\pi}{(1-\pi)\psi_\alpha}$

  $\Rightarrow$ denote probability of following paid recommendations by ψ_η

Influencer Decision

- Recall influencer's utility: $u_{\mathcal{I}} = (1-\alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - v_0]$
- Three influencer types, accept endorsement iff
  - $\bar{s}: \quad (1-\alpha)\kappa + \alpha[\pi v - p - v_0] \geq 0$
    $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \pi v + p + v_0}$
  - $s|v: \ (1-\alpha)\kappa + \alpha[(1-\pi)\sigma v + \pi v - p - v_0] > (1-\alpha)\kappa \geq 0$
    $\Rightarrow$ always true (interests perfectly aligned)
  - $s|0: (1-\alpha)\kappa + \alpha[(1-\sigma)\pi v - p - v_0] \geq 0$
    $\Rightarrow \alpha \leq \bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1-\sigma)\pi v + p + v_0}$

$\Rightarrow$ aligned paid recommendation probability $\psi_\alpha = \frac{\pi s(1 - F_\alpha(\bar{\alpha}_s))}{s(\pi + (1-\pi)F_\alpha(\bar{\alpha}_s)) + (1-s)F_\alpha(\bar{\alpha}_{\bar{s}})}$

Comparative Statics

Influencer

- $\bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \pi v + p + v_0}$ $\quad\quad$ $\bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1-\sigma)\pi v + p + v_0}$

  $\Rightarrow$ incr. in κ, π, v decr. in p, v_0 $\quad\quad$ incr. in κ, π, v decr. in p, v_0, σ
- $\psi_\alpha \equiv \frac{\pi s(1-F_\alpha(\bar{\alpha}_s))}{s(\pi+(1-\pi)F_\alpha(\bar{\alpha}_s))+(1-s)F_\alpha(\bar{\alpha}_{\bar{s}})}$

  $\Rightarrow$ incr. in s, p, v_0, σ decr. in κ, v and $\bar{\alpha}$ ambiguous in π

Consumer

- $\bar{\sigma} \equiv \frac{v_0 + p - \pi v}{(1-\pi)v\psi_\alpha}$

  $\Rightarrow$ incr. in κ decr. in ψ_α, s, σ

Firm

- $\bar{\kappa}_H \equiv \rho[\psi_\eta(1-s)F_\alpha(\bar{\alpha}_{\bar{s}}) - (1-\psi_\eta)s](p-c)$

  $\Rightarrow$ incr. in ρ decr. in c

Example: HML-Equilibrium

Parameters

- $v = 6,\ v_0 = 2,\ p = 3,\ \kappa = 0.5,\ \pi = 0.8$
- $\rho = 0.7,\ s = 0.2,\ \sigma = 0.8$
- $\alpha \sim U[0,1] \quad \sigma_\eta = \eta + (1-\eta)\sigma,\ \eta \sim U[0,1]$

Equilibrium Quantities

- $\bar{\alpha}_{\bar{s}} \approx 0.71 \quad \bar{\alpha}_s \approx 0.11$

  $\Rightarrow \psi_\alpha \approx 0.19$
- $\bar{\sigma} \approx 0.86$

  $\Rightarrow \psi_\eta \approx 0.69$
- $\bar{\kappa}_H \approx 0.7$

How to best visualize Examples?

The (first) ML-Endorsement Equilibrium

- Two (mutually exclusive) equilibria classes dependent on $\sigma v - p \gtrless v_0$

  $\Rightarrow$ focus (first) on $\sigma v - p > v_0$ [genuine recommendation for M-types]

ML-Endorsement Equilibrium (Aligned Influencers)

For intermediate κ there exists an equilibrium with endorsements in which

(a) Only firm types $q = M, L$ decide to use endorsements at cost κ.

(b) There exist thresholds $\bar{\alpha}_s$ and $\bar{\alpha}_{\bar{s}}$ such that an influencer

  i. with $\alpha \leq \bar{\alpha}_{\bar{s}}$ endorses the product unseen.

  ii. with $\alpha \leq \bar{\alpha}_s$ endorses any product and $\alpha > \bar{\alpha}_s$ only $v_\iota = v$ products.

(c) All consumers follow genuine recommendations.

(d) Sufficiently parasocial consumers ($\sigma_\eta > \bar{\sigma}$) also follow endorsements.

Firm Decision

- Firm endorses if costs are sufficiently small $\kappa \leq (D_E^q - D_0^q)(p-c)$

  $\Rightarrow$ effective demand with (D_E^q) and without (D_0^q) endorsements depends on v_ι

- Denote share of parasocial customers ($\sigma_\eta > \bar{\sigma}$) by ψ_η

v_ι	D_0	D_E
V	s	$\rho(1-s)\psi_\eta F_\alpha(\bar{\alpha}_{\bar{s}}) - \rho s(1-\psi_\eta) + s$
0	0	$\rho(1-s)\psi_\eta F_\alpha(\bar{\alpha}_{\bar{s}}) + \rho s \psi_\eta F_\alpha(\bar{\alpha}_s)$

- Note that absolute gains ($D_E^q - D_0^q$) decrease with q_ι (and hence with q)

  $\Rightarrow$ H-type firm not endorsing iff $\kappa > \rho[\psi_\eta(1-s)F_\alpha(\bar{\alpha}_{\bar{s}}) - (1-\psi_\eta)s](p-c)$

  $\Rightarrow$ others endorse iff

$$\kappa \leq \bar{\kappa}_M \equiv \rho[\psi_\eta(1-s)F_\alpha(\bar{\alpha}_{\bar{s}}) - \pi(1-\psi_\eta)s + (1-\pi)\psi_\eta s F_\alpha(\bar{\alpha}_s)](p-c)$$

Consumer Decision

- Reminder: Consumers can differentiate paid from genuine recommendations

  $\Rightarrow$ paid ones can still be **aligned** (i.e., iff $v_\iota = v$) for high α $\Rightarrow$ denote prob. ψ_α
- Expected value from unaligned recommendations: $\frac{\pi(1-\pi)}{1-\pi^2}v = \frac{\pi}{1+\pi}v$
- Genuine recommendations always aligned (Benchmark) $\Rightarrow$ always followed
- Follow paid recommendation iff v_0 or p sufficiently small and

  $$\mathbb{E}[v_\theta | r^p] = \psi_\alpha \sigma v + (1-\psi_\alpha)\frac{\pi}{1+\pi}v \geq v_0 + p$$

  $\Rightarrow$ satisfied for sufficiently parasocial consumers $\Rightarrow \sigma_\eta \geq \bar{\sigma} \equiv \frac{v_0+p}{\psi_\alpha v} - \frac{\pi(1-\psi_\alpha)}{(1+\pi)\psi_\alpha}$

  $\Rightarrow$ denote probability of following paid recommendations by ψ_η

Influencer Decision

- Recall influencer's utility: $u_{\mathcal{I}} = (1-\alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - v_0]$
- Three influencer types, accept endorsement iff
  - $\bar{s}: \quad (1-\alpha)\kappa + \alpha[\frac{\pi}{1+\pi}v - p - v_0] \geq 0$
    $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \frac{\pi}{1+\pi}v + p + v_0}$
  - $s|v: \ (1-\alpha)\kappa \geq 0$
    $\Rightarrow$ always true (interests perfectly aligned as long as $\sigma v - p \geq v_0$)
  - $s|0: (1-\alpha)\kappa + \alpha[(1-\sigma)\pi v - p - v_0] \geq 0$
    $\Rightarrow \alpha \leq \bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1-\sigma)\pi v + p + v_0}$

$\Rightarrow$ aligned paid recommendation prob. $\psi_\alpha = \frac{\pi s(1-F_\alpha(\bar{\alpha}_s))}{s(\pi + F_\alpha(\bar{\alpha}_s)) + (1+\pi)(1-s)F_\alpha(\bar{\alpha}_{\bar{s}})}$

Comparative Statics

Influencer

- $\bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \frac{\pi}{1+\pi} v + p + v_0}$ $\bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1-\sigma)\pi v + p + v_0}$

  $\Rightarrow$ incr. in κ, π, v decr. in p, v_0 incr. in κ, π, v decr. in p, v_0, σ

- $\psi_\alpha = \frac{\pi s(1 - F_\alpha(\bar{\alpha}_s))}{s(\pi + F_\alpha(\bar{\alpha}_s)) + (1+\pi)(1-s)F_\alpha(\bar{\alpha}_{\bar{s}})}$

  $\Rightarrow$ incr. in s, p, v_0, σ decr. in κ, v and $\bar{\alpha}$ ambiguous in π

Consumer

- $\bar{\sigma} \equiv \frac{v_0 + p}{\psi_\alpha v} - \frac{\pi(1-\psi_\alpha)}{(1+\pi)\psi_\alpha}$

  $\Rightarrow$ incr. in κ decr. in ψ_α, s, σ

Firm

- $\bar{\kappa}_H \equiv \rho[\psi_\eta(1-s)F_\alpha(\bar{\alpha}_{\bar{s}}) - (1-\psi_\eta)s](p-c)$

  $\Rightarrow$ incr. in ρ decr. in c

ML-Equilibrium vs HML-Equilibrium

Influencer

- Uninformed influencer less likely to endorse: $\bar{\alpha}_{\bar{s}}^{ML} < \bar{\alpha}_{\bar{s}}^{HML}$
- Informed influencer unchanged: $\bar{\alpha}_{s}^{ML} = \bar{\alpha}_{s}^{HML}$

Consumer

- More parasociality needed to follow endorsements: $\bar{\sigma}^{ML} > \bar{\sigma}^{HML}$
- Value of genuine recommendation increases

Firm

- Changes in demand and threshold only through $\bar{\alpha}, \bar{\sigma}$
- Anticipating only ML-endorsements reduces profitability: $\bar{\kappa}_H^{ML} < \bar{\kappa}_H^{HML}$

$\Rightarrow$ since $\bar{\kappa}_H^{ML} < \bar{\kappa}_H^{HML}$ $\rightarrow$ multiplicity of HML/ML equilibrium

Example: ML-Equilibrium

Parameters

- $v = 8, \ v_0 = 1, \ p = 4, \ \kappa = 0.2, \ \pi = 0.6$
- $\rho = 0.9, \ s = 0.3, \ \sigma = 0.7$
- $\alpha \sim U[0,1] \quad \sigma_\eta = \eta + (1-\eta)\sigma, \ \eta \sim U[0,1]$

Equilibrium Quantities

- $\bar{\alpha}_{\bar{s}} \approx 0.05 \quad \bar{\alpha}_s \approx 0.03$

  $\Rightarrow \psi_\alpha \approx 0.72$
- $\bar{\sigma} \approx 0.72$

  $\Rightarrow \psi_\eta \approx 0.93$
- $\bar{\kappa}_H \approx 0.04 \quad \bar{\kappa}_M \approx 0.11$

How to best visualize Examples?

The other ML-Endorsement Equilibrium

- Two (mutually exclusive) equilibria classes dependent on $\sigma v - p \gtrless v_0$

$\Rightarrow$ focus now on $\sigma v - p > v_0$ [some $v_\iota = v$ get not recommended]

ML-Endorsement Equilibrium (Non-Aligned Influencers)

For intermediate κ there exists an equilibrium with endorsements in which

(a) Only firm types $q = M, L$ decide to use endorsements at cost κ.

(b) There exist thresholds $\bar{\alpha}_0$, $\bar{\alpha}_v$ and $\bar{\alpha}_{\bar{s}}$ such that an influencer

  i. with $\alpha \leq \bar{\alpha}_{\bar{s}}$ endorses the product unseen.

  ii. with $\alpha \leq \bar{\alpha}_0$ endorses any product and $\alpha < \bar{\alpha}_v$ only $v_\iota = v$ products.

(c) All consumers follow genuine recommendations.

(d) Sufficiently parasocial consumers ($\sigma_\eta > \bar{\sigma}$) also follow endorsements.

Influencer Decision

- Similar to before, but with major difference at $v_\iota = v$

  $\Rightarrow$ additional info of endorsement offer now harmful

- Three influencer types, accept endorsement iff
  - $\bar{s}: \ \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \frac{\pi}{1+\pi} v + p + v_0}$ [unchanged] $\quad s|0: \ \bar{\alpha}_0 \equiv \frac{\kappa}{\kappa - (1-\sigma)\pi v + p + v_0}$ [unchanged]
  - $s|v: \ (1-\alpha)\kappa + \alpha[\sigma v - p - v_0] \geq 0$

    $\Rightarrow$ since $\sigma v - p < v_0$, now $\alpha < \bar{\alpha}_v \equiv \frac{\kappa}{\kappa - \sigma v + p + v_0}$

  $\Rightarrow$ One can show $\bar{\alpha}_v > \bar{\alpha}_0$ and $\bar{\alpha}_{\bar{s}} > \bar{\alpha}_0$

- Modified aligned paid recommendation probability $\psi_\alpha = \frac{s(F_\alpha(\bar{\alpha}_v) - F_\alpha(\bar{\alpha}_0))}{sF_\alpha(\bar{\alpha}_v) + (1-s)F_\alpha(\bar{\alpha}_{\bar{s}})}$

  $\Rightarrow$ Otherwise consumer problem unchanged: $\bar{\sigma} \equiv \frac{v_0 + p}{\psi_\alpha v} - \frac{\pi(1-\psi_\alpha)}{(1+\pi)\psi_\alpha}$

Firm Decision

- Firm endorses if costs are sufficiently small $\kappa \leq (D_E^q - D_0^q)(p - c)$

  $\Rightarrow$ effective demand with (D_E^q) and without (D_0^q) endorsements depends on v_ι

- Adjust endorsement demand D_E^q for abstaining influencer ($\alpha > \bar{\alpha}_v$)

v_ι	D_0	D_E
V	s	$\rho(1-s)\psi_\eta F_\alpha(\bar{\alpha}_{\bar{s}}) - \rho s(1-\psi_\eta)F_\alpha(\bar{\alpha}_v) - \rho s(1 - F_\alpha(\bar{\alpha}_v)) + s$
0	0	$\rho(1-s)\psi_\eta F_\alpha(\bar{\alpha}_{\bar{s}}) + \rho s \psi_\eta F_\alpha(\bar{\alpha}_s)$

- Note that absolute gains ($D_E^q - D_0^q$) decrease with q_ι (and hence with q)

  $\Rightarrow$ adjust firm-type thresholds $\bar{\kappa}_H$ and $\bar{\kappa}_M$ accordingly

Non-Existence of L-Endorsement Equilibria

Non-Existence of L-Endorsement Equilibria

In the baseline, there cannot exist an equilibrium in which only the L-type uses endorsements while the other types rely solely on genuine recommendations.

- Equilibrium possible from firm side with sufficiently high κ
- Paid recommendations accepted by sufficiently selfish influencer (low $\bar{\alpha}$)
- paid recommendation credibly signals $q = L \quad \Rightarrow v_\theta = 0$

  $\Rightarrow$ no paid recommendation followed by consumers

  $\Rightarrow$ no reason to endorse for L-type firm

A Sketch of the mL Mixed Endorsement Equilibrium

Mixed Endorsement Equilibria

In any equliibrium, at most one firm finds it optimal to mix at the same time.

- H-firm never endorses, L-firm always, M-firm with prob. ε

  $\Rightarrow$ expected value from endorsement: $\frac{\varepsilon\pi}{2\varepsilon\pi+1-\pi}v$

- Influencer decision: $\bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa-\frac{\varepsilon\pi}{2\varepsilon\pi+1-\pi}v+p+v_0}, \quad \bar{\alpha}_0 \equiv \frac{\kappa}{\kappa-(1-\sigma)\frac{\varepsilon\pi}{\varepsilon\pi+1-\pi}v+p+v_0}$

  $\Rightarrow$ aligned endorsement prob. : $\psi_\alpha = \frac{\varepsilon\pi s(1-F_\alpha(\bar{\alpha}_s))}{\varepsilon\pi s+(\varepsilon\pi+1-\pi)sF_\alpha(\bar{\alpha}_s)+(2\varepsilon\pi+1-\pi)(1-s)F_\alpha(\bar{\alpha}_{\bar{s}})}$

- Consumer decision: $\psi_\alpha\sigma + (1-\psi_\alpha)\frac{\varepsilon\pi}{2\varepsilon\pi+1-\pi} \geq \frac{v_0-p}{v} \quad \Rightarrow \bar{\sigma} \equiv \frac{v_0+p}{\psi_\alpha v} - \frac{\varepsilon\pi(1-\psi_\alpha)}{(2\varepsilon\pi+1-\pi)\psi_\alpha}$

- Firm decision: choose ε such that $\kappa = \bar{\kappa}_M(\varepsilon)$

What's ahead

- Influencer indirect utility conditional on following recommendation (status quo)?
- Continuous vs. binary parasocial state space?
- Characterize the prob. 0 mixed endorsement equilibria?

- ML-Endorsement Equilibrium
- Comparative Statics and Welfare/CS Analysis

- Introduce product categories and influencer "specialization" $\zeta \in K$

  $\Rightarrow$ affects σ and possibly ω_ζ $\rightarrow$ allows parasociality as spillover effects
- Endogenize ω (firm choice) $\rightarrow$ limited signalling
- Costly quality investigation for unknowing influencer

That’s All Folks!

If you have questions, comments or feedback, don't hesitate to contact me or stop by.

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Appendix: Influencer Decision with unconditional utility

- Recall influencer's utility: $u_{\mathcal{I}} = (1-\alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - v_0]$
- Three influencer types, accept endorsement iff
  - $\bar{s}: \quad (1-\alpha)\kappa + \alpha\psi_\eta[\pi v - p - v_0] \geq 0$
    $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \psi_\eta(\pi v - p - v_0)}$
  - $s|v: \ (1-\alpha)\kappa + \alpha\psi_\eta[\sigma v + (1-\sigma)\pi v - p - v_0] \geq \alpha[\sigma v + (1-\sigma)\pi v - p - v_0]$
    $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}}^{v} \equiv \frac{\kappa}{\kappa + (1-\psi_\eta)(\sigma v + (1-\sigma)\pi v - p - v_0)}$
  - $s|0: (1-\alpha)\kappa + \alpha\psi_\eta[(1-\sigma)\pi v - p - v_0] \geq 0$
    $\Rightarrow \alpha \leq \bar{\alpha}_{s}^{0} \equiv \frac{\kappa}{\kappa - \psi_\eta((1-\sigma)\pi v - p - v_0)}$

$\Rightarrow$ one can show: $\bar{\alpha}_{\bar{s}}^{v} > \bar{\alpha}_{\bar{s}} > \bar{\alpha}_{s}^{0}$